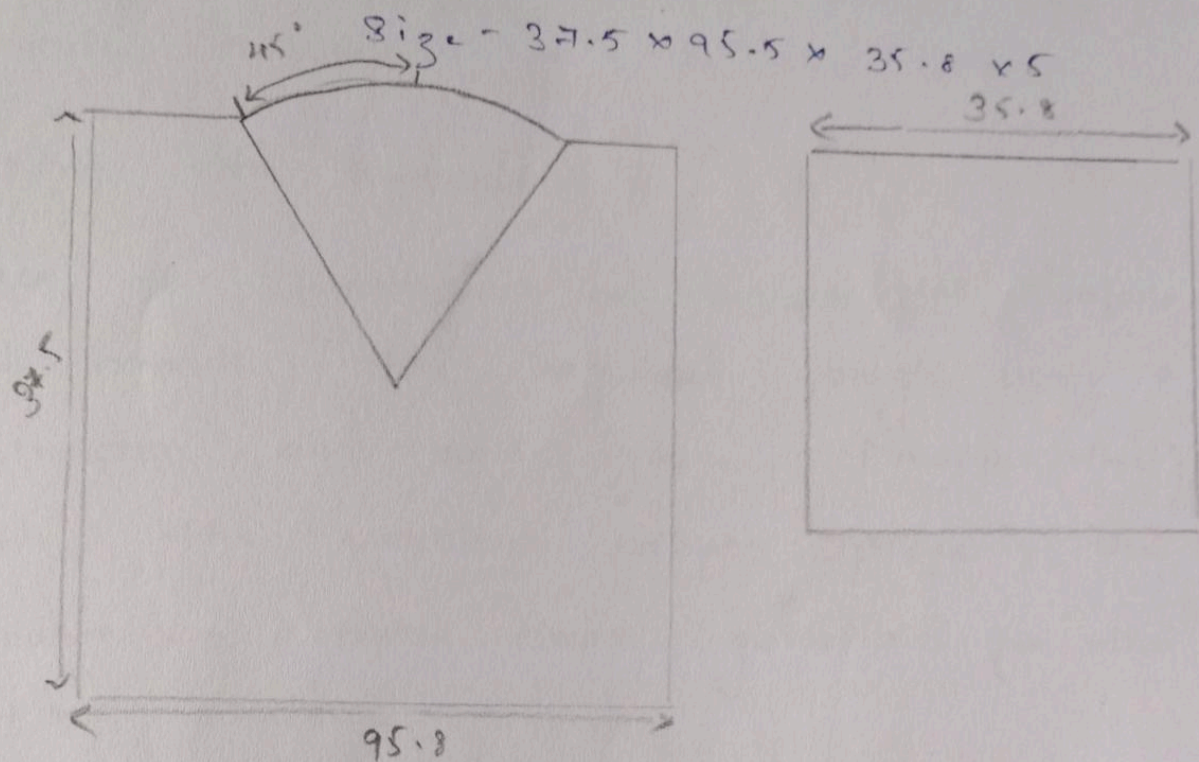


Aim = Angle Cutting

Experiment : To make job as per drawing material

- M/s



Tools required

1. Shaper machine
2. Angle gauge or protractor
3. Measuring tape or vernier ~~cal~~ caliper
4. Marking tool
5. Vice or clamps
6. File
7. Protective gear

Procedure

1) Study the drawing

Understand the dimensions and required angles

2) Preparation of the Workstation

Inspect the shaper machine to ensure it is clean, functional and well lubricated. Verify that the cutting tool is sharp and securely mounted in the tool holder.

3) Making the Workpiece

Place the workpiece on a flat surface and mark the required angle using a protractor or angle gauge. Measure the job with the vernier caliper properly. Use a scriber to draw clean guidelines for the cutting process.

4) Securing the workpiece

Fix the workpiece securely on the shaper machine table using a vice. Ensure the marked angle is slightly aligned properly with the machine's cutting path.

5) Adjusting the shaper table.

Tilt the shaper table to the required angle as specified in the drawing. Use an angle gauge to confirm the accuracy of the table adjustments.

6) Setting the Cutting tool

Align the cutting tool with the marked line on the workpiece. Set the tool height and position for smooth and precise cutting operation.

7) Cutting operation

Start the shaper machine and gradually feed the workpiece into the cutting tool. Monitor the cutting process to ensure the angle is being cut accordingly and accurately. Perform multiple passes if required to achieve the desired depth and finish.

8) Measurement and Inspection

After completing the cut, stop the machine and remove the workpiece. Measure the angle using a protractor to verify its accuracy, compare the finished job with the drawings specification.

9) Finishing

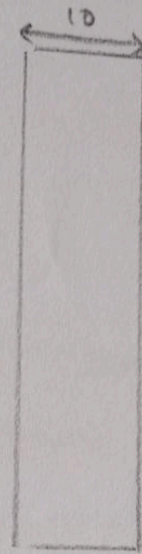
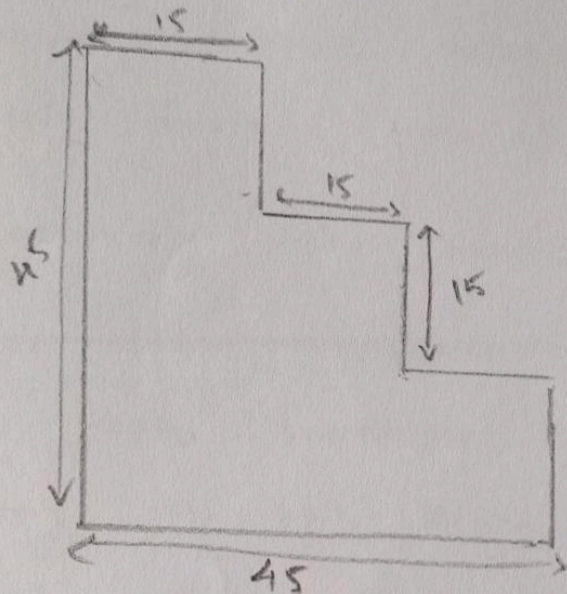
Use a file to smoothen any rough edges or burrs on the cut surface. Polish the surface with sandpaper for a clean and professional finish.

Aim : Step Cutting

Experiment : To make a job as per drawing

Materials : M/s

Size : $45 \times 45 \times 10$



Tools required

1. Shaper machine
2. Vernier caliper
3. Scriber or marking tool
4. Measuring tape or scale
5. Vio
6. File

Procedure

1) Preparation of the workstation

Inspect of the ~~workstation~~ shaper machine to ensure it is clean and functional. Check the sharpness of the cutting tool and secure it firmly in the tool holder. Set the stroke length according to the size of the workpiece.

2) Mark the workpiece

Place the workpiece on a flat surface. Use a scriber and measuring tools to mark the width and depth of each step based on the drawing. Ensure markings are clean, precise and visible during machining.

3) Securing the workpiece

~~Place the workpiece on a flat surface.~~

Fix the workpiece on the shaper machine table using a vice. Align the first marked step with the cutting tool path for accuracy.

4) Cutting the first step

Adjust the shaper machine to align the cutting tool with the first step marking. Begin machining to the specified depth with a depth gauge after completing the cut to ensure accuracy.

5) Cutting subsequent steps

Reposition the workpiece or adjust the table height for the next step. Repeat the cutting process for each step. Verifying dimensions after each cut.

6) Measure and Verify

After completing all steps, stop the machine and remove the workpiece. Measure the dimension of each step using a vernier caliper or depth gauge. Compare the measurement with the drawing to ensure accuracy.

7) Finishing the Workpiece

Smooth the edges and surfaces ~~with sand~~ of each step using a file. Polish the entire surface with ~~sand~~ sandpaper to remove burrs and achieve a clean finish.

8) Clean the Work

Clean the workpiece and remove debris from the shaper machine and worktable. Reinspect the final workpiece to confirm it matches the technical drawing.

Aim: To cut surface of a material to make it smooth.

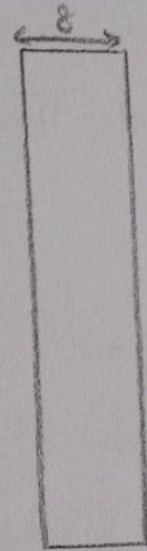
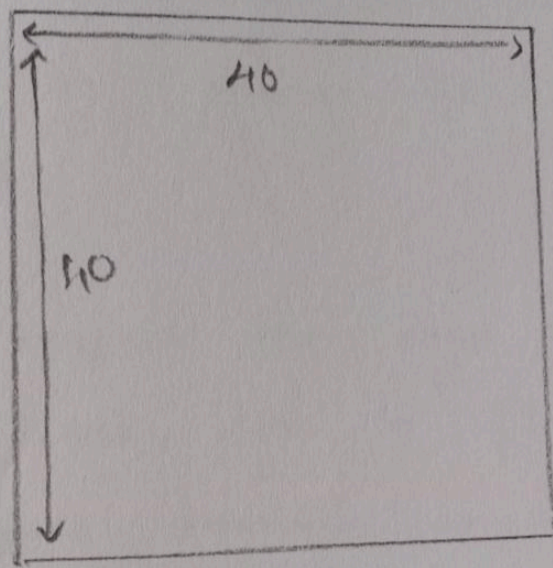
Exp: To make a job as per drawing

Materials - M/s

Size - $40 \times 40 \times 8$

Tools required

- Measuring tape or ruler
- Surface gauge or marking tool
- Cutting tool
- File



Procedure

1. Operating

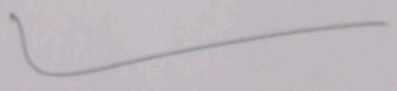
Secure the material on a stable work station using clamps or vice to prevent movement during cutting. Inspect the cutting tool to ensure it is suitable for the material and in proper working condition.

2) Surfacing

Examine the material for irregularities or unevenness. Mark the surface to be cut using a surface gauge or marking tool ensuring precise measurement. Use the appropriate cutting tool to cut the material along the marked lines. Maintain control and steady movement during the cutting process to avoid deviation.

3) Finishing

After cutting inspect the edges and surface for sharpness or burrs. Use a file to smooth out rough edges and remove any burrs. Polish the surface with sandpaper to achieve a smooth and finished appearance. Clean the material and ensure the cut matches the required dimensions and specification.



Aim → Ring cutting

E x P → To make a job as per drawing ~~measurement~~



Tools requirement →

- (i) Scrubber or masking tool.
- (ii) Shaper machine
- (iii) Measuring tape or scale
- (iv) Vice
- v



Procedure →

- (1) Study the drawing →

Understand the dimension and required angles

- (2) Preparation of the workpiece →

Inspect the shaper machine to ensure it is clean, functional and well lubricated. Verify that the cutting tool is sharp and securely mounted in the tool holder.

- (3) Making the workpiece →

Place the workpiece on a flat surface and mark the required angle using a protractor or angle gauge measure the job ~~with the~~. Use a scriber to draw clear guidelines for the cutting process.

④ Securing the Work piece →

Fix the Workpiece securely on the Shaper machine table using a vice.
Ensure the marked dots is aligned properly with the machines cutting path.

⑤ Adjusting the Shaper Tools table →

~~Tilt the shaper table to the required angle as specified in the drawing~~

⑥ Setting the cutting tool →

Align the cutting tool with the marked line on the workpiece. Set the tools height and position for smooth and precise cutting operation.

⑦ Cutting operation →

Start the shaper machine and gradually feed the workpiece into the cutting tool. Monitor the cutting process to ensure the angle is being cut accordingly and accurately perform multiple passes if required to achieve the desired depth and finish.

⑧ Measurement and inspection →

After completion its cut, stop the machine and remove the workpiece. Measure the dot to verify its accuracy. Compare the finish form with the drawings specification.

⑨ Finishing →

g) Finishing

Use a file to smoothen any rough edges on burrs on the cut surface. Polish the surface with sandpaper for a clean and professional finish.